

Those Flowing Locks - Creating Large Scale Water Hair for *Ruby Gillman Teenage Kraken*

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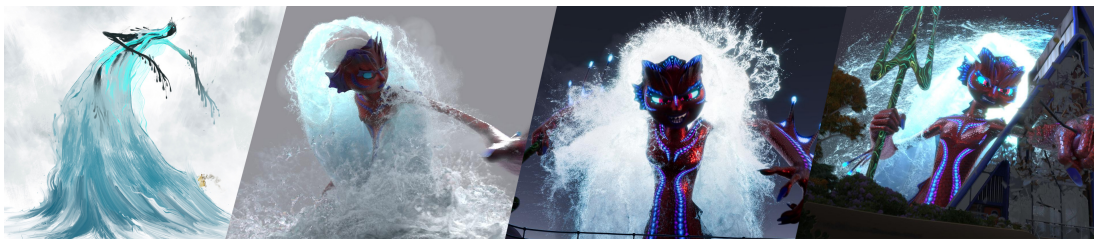


Figure 1: Concept Art, Look Development Test and Production Stills of the Titan Mermaid

ABSTRACT

One of the most unique features of the villain *Nerissa* in DreamWorks Animation's *Ruby Gillman Teenage Kraken* is her hair. In the climatic battle, *Nerissa* transformed into a 150-foot-tall Titan Mermaid, whose hair is made of flowing water drawn from the surrounding ocean. While other films have created characters based on water, the sheer scale and the required action of the character interaction with the ocean, and the number of shots involved added significant challenges to the work. Working closely with multiple departments across the pipeline, a robust rig for the water hair was developed based on a combination of procedural and simulated elements. Furthermore, a Fluid Implicit Particle (FLIP) simulation is used to simulate the character's interaction with the surrounding ocean. Finally, an additional level set fillet generation technique was used to seamlessly blend the ocean mesh with the water hair geometry.

CCS CONCEPTS

• Computing methodologies → Procedural Animation; Simulation.

KEYWORDS

procedural animation, liquid, simulation

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1 ARTISTIC DIRECTION

The development of the element started with rough paintings from our art department (Figures 2). To convey the scale and mass of the huge character, we took inspiration from the elements we find in nature associated with large bodies of water and also some visual elements of the shape language of the film.

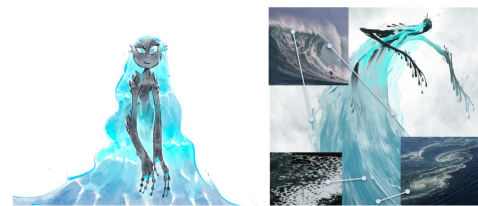


Figure 2: Concept Painting of the Titan Mermaid

2 BASE HAIR GEOMETRY ANIMATION

We started with a rigged, low-poly "hair shell" model, which consists of several tubular structures that represented bundles of hair. After the character's performance was approved, the CFX department further refined the hair shape. The final hair geometry was layered on top of a softbody tube simulation. This both stabilized

the performance of the hair mesh and allowed the motion of the hair geometry to be somewhat divorced from the activity of the character's overall motion. In the softbody simulation, the base of each tube includes a constraint to a ray projection against the water level to ensure a consistent contact point as they slide across the water surface. Other considerations included a high air resistance to slow recovery speed, which helps emphasize the scale and mass of the water hair.

3 GENERATING THE WATER HAIR SURFACE

Given the hair geometry, FX artists took over. A modular setup was developed such that consistent look and feel can be quickly achieved out of the box by different artists working in multiple shots simultaneously.

3.1 Procedural wave displacement

Instances of Tessendorf's FFT-based deep water waves spectra were scattered onto the hair geometry in parametric space. The resulting displacements from the ocean spectra naturally stretch and compress as the underlying hair geometry deforms. The wave direction was aligned with the parametric direction of the hair geometry toward the scalp. Combining this with offsetting the sampling position into the ocean spectra over time, we were able to achieve the look of waves traveling the length of the hair toward her head. To complete the look, an emissive volume was generated from the hair geometry. The color and intensity was controllable along the length of the hair through a color ramp.

3.2 Ocean interaction and integration

For shots where the area of the ocean connecting with the water hair is visible in camera, a FLIP container was created at the base of the hair to simulate agitated water elements. Areas of negative divergence were added in the FLIP simulation around the base of the hair to drive fluid flow toward the Titan. Additional upward "suction" forces were added to pull fluid particles near the hair geometry toward the hair above the water line. The whitewater generated were affected by a number "repellers," which are a sparse set of points advected by the liquid velocity. The repelling forces were calibrated to create fractals of circular negative space in the resulting foam pattern which echoes the shape language for white-water and foam elements through out the movie.

3.3 VDB fillet

In order to seamlessly integrate the procedurally generated hair geometry with the FLIP-simulated water surface mesh, we first turned both meshes into level sets. To combine two level sets together, we implemented a VDB [Museth 2013] blending method proposed in [Allen et al. 2007]. A regular CSG union operation takes the minimum of two signed distance functions, which often introduces hard edges and cut-offs in the region where the two level sets intersect.

The algorithm proposed by [Allen et al. 2007] offsets the value from a regular union operation so that the two surfaces blend together in a smoother manner. Three parameters control the shape of the fillet: α (controlling the radius of the blending-region), γ (acting as a multiplier to the offset function), and β (controlling

how quickly the offset function goes to zero). If d_A and d_B denote the signed distance values of the two level sets, the algorithm computes a new distance d according to:

$$m = \text{clamp}\left(\frac{\alpha - d_A}{\alpha}, 0, 1\right) \cdot \text{clamp}\left(\frac{\alpha - d_B}{\alpha}, 0, 1\right), \quad (1)$$

$$d = \min(d_A, d_B) - \gamma m^\beta. \quad (2)$$

Note that the formulation assumes a level set convention of negative-inside/ positive-outside sign convention. We find that the algorithm is stable and robust. It is easy to find parameter values that smoothly blend two surfaces together by attenuating the exponent parameter β . A drawback of the algorithm is its requirement to extrude the exterior band of the input level sets, which needs to be sufficiently large for the parameter α to have an effect. Yet another approach to solve this problem is by moving the result of a union operation in the direction of the principal curvature [Museth et al. 2002].

4 SECONDARY ELEMENTS

Several additional elements were added to further enhance the aesthetics and the sense of scale of Nerissa. A foam element with circular negative space was added near the bottom portion of the hair. This element connects the based of the procedurally generated hair geometry to actively churning water in the surrounding ocean, creating whitewater and foam. Two types of spray elements were added. The "local" spray was simulated in the reference 3D space and then wrap-deformed to the animated position. The "world" spray was simulated in world space based on the motion of the animated model due to sudden acceleration in the character's performance. Both types of sprays consist of a particle and volume layer. Additional world-space sprays were added when other collision objects, such as Nerissa's arms and body interacted with the water hair as well.

5 CONCLUSIONS

We presented a robust and flexible system to create the 150-foot monster's hair made entirely of water. In all, the water hair element was successfully implemented in over 60 shots across two action sequences where the Titan Mermaid battled multiple giant Krakens in the middle of the ocean during the climatic action sequences of the film.

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REFERENCES

- Chris Allen, Jonathan M. Cohen, Doug Bloom, Daniel P. Ferreira, and Sho Hasegawa. 2007. Levelssets in production: Spider-man 3. In *ACM SIGGRAPH 2007 Sketches*. ACM, ACM Press / ACM SIGGRAPH, 29–es. <https://doi.org/10.1145/1278780.1278815>
- Ken Museth. 2013. VDB: High-resolution Sparse Volumes with Dynamic Topology. *ACM Transactions on Graphics* 32, 3 (June 2013). <https://doi.org/10.1145/2487228.2487235>
- Ken Museth, E. David Breen, Ross T. Whitaker, and Alan H. Barr. 2002. Level set surface editing operators. *ACM Transactions on Graphics* 21, 3 (July 2002), 330–338. <https://doi.org/10.1145/566654.566585>